Birthdays, Derangements, and a Little Calculus

50 minutes • 50 points • Instructor key

Grading note. Subpart scores sum to the problem totals. Award method credit for correct reasoning after an earlier error; do not deduct twice for the same arithmetic mistake. Suggested working time is 45 minutes, leaving 5 minutes of slack.

Instructions. Show your reasoning. Give exact fractions where a number is requested; no calculator is needed. You may use $\ln 2 \approx 0.693$ and $1 + 2 + \cdots + (n-1) = n(n-1)/2$ without proof. The suggested times leave 5 minutes of slack.

1. The calculus toolkit

10 points · about 8 minutes

(a) Write the Taylor series of e^x about $x = 0$: give the terms through x^3 and the general term. Then write $1/e$ as an infinite series of numbers. (3 points)

Solution.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad \frac{1}{e} = e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots .$$

Credit: Terms through x^3: 1; general term: 1; series for $1/e$: 1.

(b) Let $f(x) = \ln(1 - x)$. Find $f(0)$ and $f'(0)$, and write the tangent-line approximation to $f(x)$ for x near 0. (3 points)

Solution. $f(0) = \ln 1 = 0$. By the chain rule $f'(x) = -1/(1-x)$, so $f'(0) = -1$. The tangent line at 0 is $f(0) + f'(0)x$, giving $\ln(1-x) \approx -x$ for small x.

Credit: $f(0)$: 1; $f'(0)$ with the chain-rule sign: 1; approximation: 1.

(c) Find $\lim_{n\to\infty} n \ln\!\left(1 - \frac{1}{n}\right)$, and use it to find $\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n$. Justify the first limit (the definition of $f'(0)$ from part (b), or L'Hôpital's rule, will do). (4 points)

Solution. Put $x = 1/n \to 0$. Since $f(0) = 0$,

$$n \ln\!\left(1 - \frac{1}{n}\right) = \frac{f(x) - f(0)}{x} \longrightarrow f'(0) = -1.$$

Because $(1 - 1/n)^n = \exp\!\big(n \ln(1 - 1/n)\big)$ and exp is continuous, the second limit is $e^{-1} = 1/e$.

Credit: Rewriting as a difference quotient or a 0/0 form: 1; first limit -1 with justification: 1; taking logarithms/exponentials: 1; answer $1/e$: 1. "$\ln(1-x) \approx -x$, so the limit is -1" without further comment earns the justification point.

2. Shared birthdays on a d-day calendar

14 points · about 13 minutes

A year has d days. People enter a room one at a time; their birthdays are independent and uniformly distributed over the d days. Let N_n be the event that the first n people all have *different* birthdays.

(a) Suppose the first k people have k different birthdays. What is the conditional probability $\mathbb{P}(N_{k+1} \mid N_k)$ that person $k+1$ misses all of them? Use the multiplication rule to write $\mathbb{P}(N_n)$ as a product (for $n \le d$). (4 points)

Solution. Given N_k, exactly k of the d equally likely days are taken, so $\mathbb{P}(N_{k+1} \mid N_k) = (d-k)/d = 1 - k/d$. Since $N_n \subset N_{n-1} \subset \cdots \subset N_1$ and $\mathbb{P}(N_1) = 1$,

$$\mathbb{P}(N_n) = \mathbb{P}(N_1) \prod_{k=1}^{n-1} \mathbb{P}(N_{k+1} \mid N_k) = \prod_{k=1}^{n-1} \left(1 - \frac{k}{d}\right).$$

Credit: Conditional probability: 2; multiplication rule: 1; product with correct range: 1. The counting form $d(d-1)\cdots(d-n+1)/d^n$ is equivalent and earns the last two points.

(b) Take the logarithm of your product and use $\ln(1-x) \approx -x$ to show that, when n is much smaller than d, (4 points)

$$\mathbb{P}(N_n) \approx \exp\!\left(-\frac{n(n-1)}{2d}\right).$$

Solution.

$$\ln \mathbb{P}(N_n) = \sum_{k=1}^{n-1} \ln\Big(1 - \frac{k}{d}\Big) \approx -\sum_{k=1}^{n-1} \frac{k}{d} = -\frac{1}{d} \cdot \frac{n(n-1)}{2},$$

where each $k/d \le n/d$ is small. Exponentiate both sides.

Credit: Logarithm of a product is a sum: 1; linear approximation applied termwise: 1; arithmetic sum: 1; exponentiating: 1.

(c) On a planet whose year has $d = 100$ days, use the approximation in (b) to find the smallest n for which a shared birthday is predicted to be more likely than not. (3 points)

Solution. We need $\exp(-n(n-1)/200) < 1/2$, that is, $n(n-1) > 200 \ln 2 \approx 138.6$. Since $12 \cdot 11 = 132$ and $13 \cdot 12 = 156$, the answer is $n = 13$.

Credit: Inequality $n(n-1) > 2d \ln 2$: 1; numerical threshold: 1; $n = 13$: 1. (The exact product gives match probability 0.4968 for $n = 12$ and 0.5572 for $n = 13$, so the prediction is correct.)

(d) Now only three people are in the room. Find the probability that all three share one birthday, *given* that at least two of them do. Simplify your answer as a function of d. (3 points)

Solution. Let S be "all three share" and $M = N_3^c$ be "at least one match." Then $S \subset M$, so $\mathbb{P}(S \cap M) = \mathbb{P}(S) = d/d^3 = 1/d^2$, while

$$\mathbb{P}(M) = 1 - \frac{(d-1)(d-2)}{d^2} = \frac{3d-2}{d^2}.$$

Hence $\mathbb{P}(S \mid M) = \dfrac{1/d^2}{(3d-2)/d^2} = \dfrac{1}{3d-2}$.

Credit: $\mathbb{P}(S)$: 1; $\mathbb{P}(M)$ by the complement: 1; conditional probability using $S \subset M$: 1.

3. The hat-check problem

16 points · about 14 minutes

Each of n guests checks a hat, and the hats are returned in a uniformly random order: all $n!$ assignments are equally likely. Let A_i be the event that guest i receives their own hat, and let p_n be the probability that *nobody* receives their own hat (the assignment is a *derangement*).

(a) For $n = 3$, list all the derangements and find p_3. (3 points)

Solution. Writing the hats received by guests $1, 2, 3$ in order, the six assignments are $123, 132, 213, 231, 312, 321$. Only 231 and 312 leave no guest with their own hat, so $p_3 = 2/6 = 1/3$.

Credit: Each derangement: 1; probability: 1.

(b) For distinct guests $i_1, \dots, i_k$, explain why $\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = (n-k)!/n!$. Then use inclusion–exclusion to show that (5 points)

$$p_n = \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Solution. If guests $i_1, \dots, i_k$ have their own hats, the other $n-k$ hats may go to the other $n-k$ guests in any of $(n-k)!$ ways, out of $n!$ equally likely assignments. There are $\binom{n}{k}$ ways to choose the k guests, so the k-th inclusion–exclusion sum is

$$\binom{n}{k}\frac{(n-k)!}{n!} = \frac{n!}{k!\,(n-k)!}\cdot\frac{(n-k)!}{n!} = \frac{1}{k!}.$$

Therefore

$$\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!}, \qquad p_n = 1 - \mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Credit: Counting the intersection: 1; $\binom{n}{k}$ terms: 1; simplification to $1/k!$: 1; alternating signs: 1; complement: 1.

(c) Evaluate p_4 as a fraction in lowest terms. (2 points)

Solution. $p_4 = 1 - 1 + \dfrac{1}{2} - \dfrac{1}{6} + \dfrac{1}{24} = \dfrac{12 - 4 + 1}{24} = \dfrac{9}{24} = \dfrac{3}{8}$.

Credit: Five correct terms: 1; 3/8: 1.

(d) What is $\lim_{n\to\infty} p_n$? The error made by truncating an alternating series with decreasing terms is at most the first omitted term. Use this to find the smallest n for which this estimate guarantees that p_n is within $1/1000$ of its limit. (4 points)

Solution. By Problem 1(a), p_n is the n-th partial sum of the series for $1/e$, so $p_n \to 1/e$. The terms $1/k!$ decrease to 0, so

$$\left|p_n - \frac{1}{e}\right| \le \frac{1}{(n+1)!}.$$

We need $(n+1)! \ge 1000$. Since $6! = 720$ and $7! = 5040$, the smallest such n is $n = 6$.

Credit: Limit $1/e$: 1; bound $1/(n+1)!$: 1; factorial comparison: 1; $n = 6$: 1. (In fact $|p_5 - 1/e| \approx 0.0012$, so $n = 6$ is also the smallest n that actually works.)

(e) For $n = 4$, find the probability that nobody receives their own hat, given that guest 1 does not. (2 points)

Solution. Let D be the derangement event. Then $D \subset A_1^c$ and $\mathbb{P}(A_1^c) = 1 - 1/4 = 3/4$, so

$$\mathbb{P}(D \mid A_1^c) = \frac{\mathbb{P}(D)}{\mathbb{P}(A_1^c)} = \frac{3/8}{3/4} = \frac{1}{2}.$$

Credit: $\mathbb{P}(A_1^c)$ and $D \subset A_1^c$: 1; quotient: 1.

4. Was the clerk careless?

10 points · about 10 minutes

A hat-check clerk is *careful* with probability 2/3 and *careless* with probability 1/3. A careful clerk returns every hat to its owner. A careless clerk returns the hats in a uniformly random order, as in Problem 3. Tonight there are $n = 3$ guests. Let L mean "the clerk is careless," let A_1 mean "guest 1 receives their own hat," and let C mean "all three guests receive their own hats."

(a) Find $\mathbb{P}(C \mid L)$ and $\mathbb{P}(C \mid L^c)$, and use the law of total probability to find $\mathbb{P}(C)$. (3 points)

Solution. Exactly one of the $3! = 6$ assignments is correct, so $\mathbb{P}(C \mid L) = 1/6$, while $\mathbb{P}(C \mid L^c) = 1$. Hence

$$\mathbb{P}(C) = \frac{1}{6} \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{1}{18} + \frac{12}{18} = \frac{13}{18}.$$

Credit: Two conditional probabilities: 1; total-probability setup: 1; result: 1.

(b) All three guests receive their own hats. Use Bayes' Rule to find the probability that the clerk was careless. (3 points)

Solution.

$$\mathbb{P}(L \mid C) = \frac{\mathbb{P}(C \mid L)\mathbb{P}(L)}{\mathbb{P}(C)} = \frac{(1/6)(1/3)}{13/18} = \frac{1/18}{13/18} = \frac{1}{13}.$$

Credit: Bayes setup with the prior in the numerator: 1; substitution (or consistent use of part (a)): 1; evaluation: 1.

(c) Suppose instead you learn only that guest 1 received their own hat. Find $\mathbb{P}(L \mid A_1)$. (2 points)

Solution. $\mathbb{P}(A_1 \mid L) = 2!/3! = 1/3$ and $\mathbb{P}(A_1 \mid L^c) = 1$, so

$$\mathbb{P}(L \mid A_1) = \frac{(1/3)(1/3)}{(1/3)(1/3) + 1 \cdot (2/3)} = \frac{1/9}{7/9} = \frac{1}{7}.$$

Credit: $\mathbb{P}(A_1 \mid L) = 1/3$: 1; Bayes evaluation: 1.

(d) Redo part (b) for a general number n of guests, and say what happens to $\mathbb{P}(L \mid C)$ as $n \to \infty$. Interpret the limit in one sentence. (2 points)

Solution. Now $\mathbb{P}(C \mid L) = 1/n!$, so

$$\mathbb{P}(L \mid C) = \frac{\frac{1}{3} \cdot \frac{1}{n!}}{\frac{1}{3} \cdot \frac{1}{n!} + \frac{2}{3}} = \frac{1}{1 + 2\,n!} \longrightarrow 0.$$

A careless clerk almost never gets a long list entirely right, so a perfect return is overwhelming evidence of care. ($n = 3$ recovers 1/13.)

Credit: Formula: 1; limit with interpretation: 1.